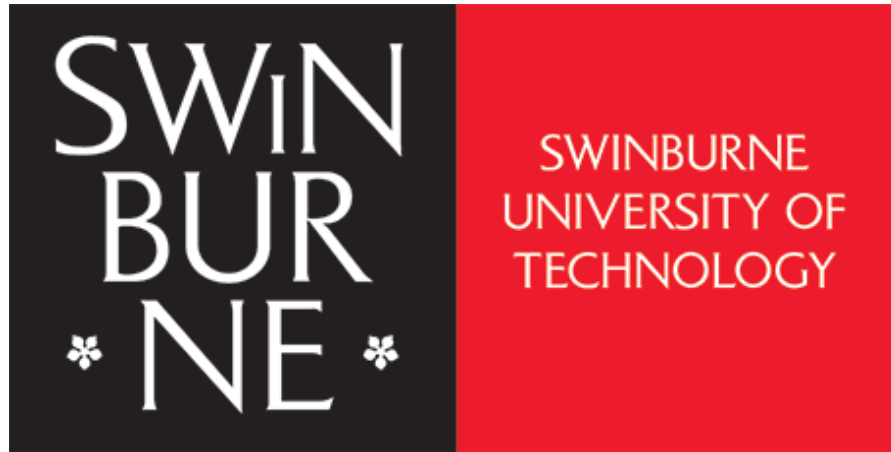




2026-HS1-COS10005-Web Development

Workshop Guide

Week 5 – Content Structure and CSS Page Layout

Aims:

- To understand the importance of designing prior to CSS implementation.
- To mark up a document using HTML5 content models for presentation using CSS.
- To learn the techniques and skills required for assignment 1.

Exercise: Structure and Present a Webpage (10 marks)

Step 1: Folder Creation

- 1.1 Create a **lab05** folder on the Mercury server and upload and test your work from there.
- 1.2 Download the file `surveyhome.html` from Canvas and use it as a template for this lab work.

Step 2: Webpage Design

- 2.1 Using the home page designed Figure 1, identify the HTML elements required to structure `surveyhome.html`. Determine if an “id” or “class” attribute is needed for the HTML elements.

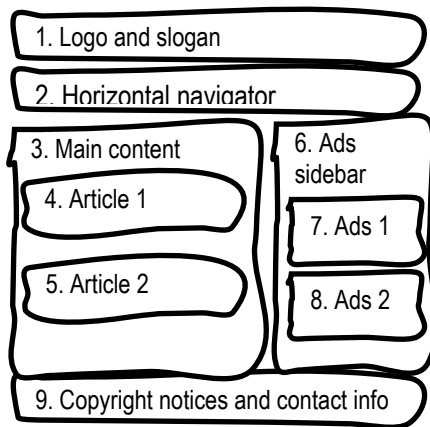


Figure 1: Mock Up of Home Page

	Element	id / class / not required
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		

Step 3: Basic HTML

Use VS Code to open and edit `surveyhome.html`.

- 3.1 Add the HTML general structure below to the document to construct the webpage.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Survey Service</title>
  <meta charset="utf-8" />
  <meta name="description" content="Web Development" >
  <meta name="keywords" content="HTML and CSS" >
  <meta name="author" content="put your name here" >
</head>
<body>
...
</body>
</html>
```

- 3.2 Structure the webpage according to the mock up presented in Figure 1 using HTML5 structural elements, e.g., `<header>`, `<nav>`, `<main>`, `<article>`, `<section>`, `<aside>`, `<footer>` and other appropriate HTML structural elements.
- 3.3 Mark up the document within each HTML structural element using appropriate HTML content elements, e.g., `<h#>`, `<p>`, `<ul>`, `<a>`, `<img>` and other appropriate elements. Specify “id” or “class” attribute as needed.

Step 4: Folder and CSS File Creation

4.1 Create a new CSS file named surveyhome.css.

4.2 Create a folder named css and save surveyhome.css in that folder

Note: A website might have more than one CSS file, so it is a good practice to store all CSS files in a separate folder. See Figure 2 for example.

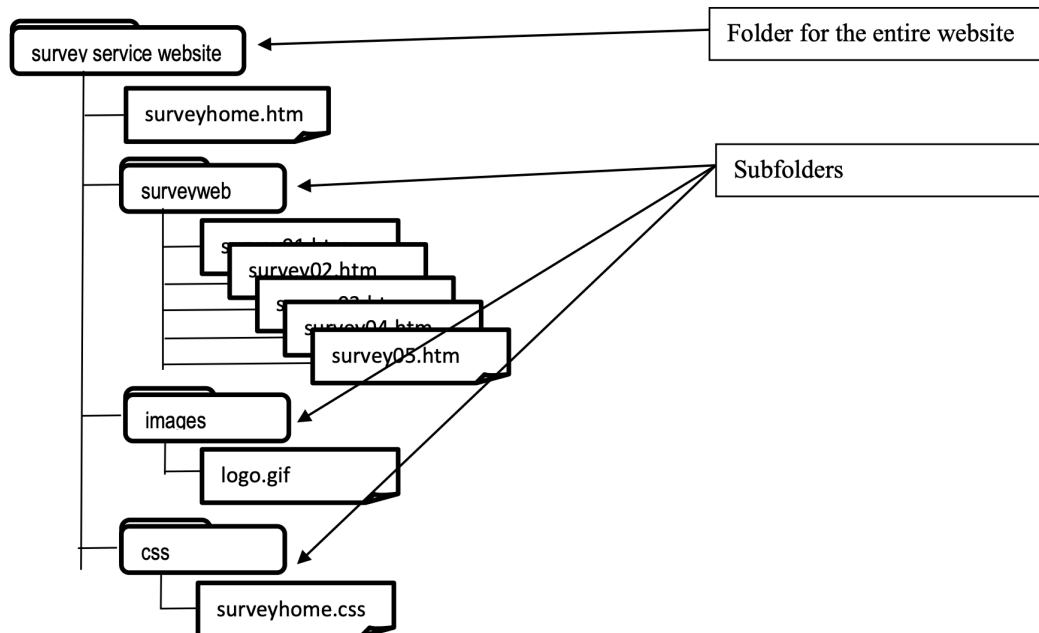


Figure 2: Directory structure

4.3 Add the following code into the head section (below the title element) of surveyhome.html. This will link surveyhome.css to surveyhome.html.

```
<link href="css/surveyhome.css" rel="stylesheet" type="text/css" />
```

Step 5: CSS Layout

Complete and apply CSS rules below applicable HTML structural elements shown in Figure 3.

Note: You might need to specify the id or the class attributes of applicable HTML elements so that they can be properly selected in the CSS file for CSS application.

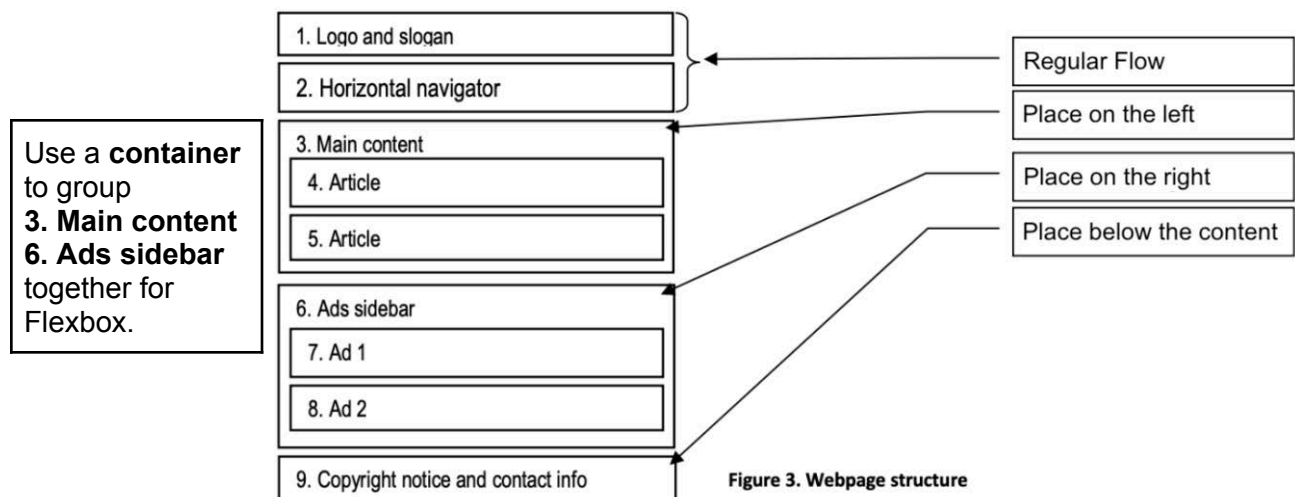


Figure 3: Webpage structure

5.1 For the layout container:

- Enable Flexbox layout
- Add spacing between elements

```
_____ { /*CSS Selector*/  
  display : _____;  
  gap : _____;  
}
```

5.2 For the `<main>` element:

- Allow the main content to take more space (Hint: Use a larger flex value for the main content. A common layout ratio is 3:1, e.g., `flex: 3`.)

```
_____ { /*CSS Selector*/  
  flex : _____;  
}
```

5.3 For the `<aside>` element:

- Place beside main content with proper width proportion
- Apply typography styling (Set font size to be 8px, font style to be italic, and color to be green)

```
_____ { /*CSS Selector*/  
  flex : _____;  
  font-size : _____;  
  font-style : _____;  
  color : _____;  
}
```

5.4 For the `<footer>` element:

- Add spacing above footer

```
_____ { /*CSS Selector*/  
  margin-top : _____;  
}
```

Step 6: CSS Web Typography and Colouring

6.1 Create and apply CSS rules to HTML elements as indicated in Figure 4.

Note: You might need to specify the **id** or the **class** attributes of applicable HTML elements so that they can be properly selected in the CSS file for CSS application.

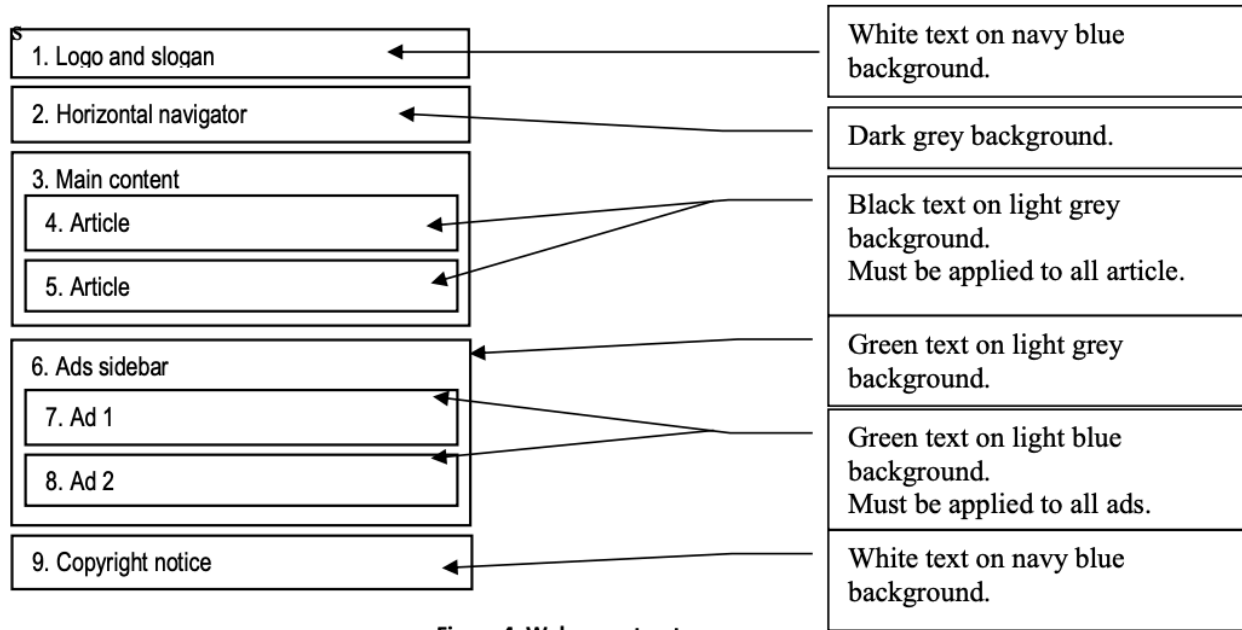


Figure 4. Webpage structure

Complete and apply the following CSS rules to match the layout and styles shown in Figure 4.

____ { /* CSS Selector */ color: ____; background-color: ____; }	____ { /* CSS Selector */ background-color: ____; }	____ { /* CSS Selector */ color: ____; background-color: ____; }
____ { /* CSS Selector */ color: ____; background-color: ____; }	____ { /* CSS Selector */ color: ____; background-color: ____; }	____ { /* CSS Selector */ color: ____; background-color: ____; }

Step 7: More CSS

Apply more CSS to surveyhome.html as instructed below

Note: You might need to specify the id or the class attributes of applicable HTML elements so that they can be properly selected in the CSS file for CSS application.

7.1 For the list (`<ul>`) element in `<nav>`:

- Display the list items in a horizontal row using Flexbox;
- Remove all bullet point symbols;
- Eliminate default padding and margin; and
- Make the list occupy the entire width of the navigation area.

```

____ { /*CSS Selector*/
display : ____;
list-style-type : ____;
padding : ____;
margin : ____;
width: ____;
}

```

7.2a For all elements in <nav>:

- Make each navigation item take equal space across the navigation bar.

```

_____ { /*CSS Selector*/
flex : _____;
}

```

7.2b For all <a> elements in <nav>:

- Display each link as a block element;
- Remove the underline from the links;
- Set the font color to white;
- Set the background color to blue;
- Set the top and bottom padding to be 0.2em;
- Set the right and left padding to be 0.6em; and
- Add a right border that is 1px, solid and white.

```

_____ { /*CSS Selector*/
display : _____;
text-align : _____;
text-decoration : _____;
color : _____;
background-color : _____;
padding : _____;
border-right : _____;
}

```

7.3 Complete and apply the following CSS rule to all the <a> elements in the <nav> part of surveyhome.html using an appropriate pseudo class selector:

- Set the background color of the <a> elements to be brown on **mouse over**.

```

_____ { /* CSS Selector */
background-color : _____;
}

```

7.4 Complete and apply the following CSS rule to all the <article> elements in surveyhome.html using an appropriate selector:

- Set the width to be automatic;
- Set the height to be automatic; and
- Set the box shadow to be 3px 3px 3px 2px #797979.

```

_____ { /* CSS Selector */
width : _____;
height : _____;
box-shadow : _____;
}

```

7.5 Complete and apply the following CSS rule to all the <h1> and <h2> elements in all the <article> elements in surveyhome.html using an appropriate selector:

- Set the text shadow to be #aaa 2px 2px 2px.

```

_____ { /* CSS Selector */
text-shadow : _____;
}

```

Step 8: Webpage Testing

8.1 Using WinSCP (or Cyberduck for mac users), upload your files, including HTML, css files and images (if any), onto Mercury. 8.2 To view the pages through http, use any Web browser and type in the following address,

`http://mercury.swin.edu.au/<your unit code>/s<your Swinburne ID>/<folder>/<filename>`

Please refer to the following examples to identify the URLs of your webpages.

Folder on Mercury Web Server	URL
~/cos10005/www/htdocs/lab05/surveyhome.html	http://mercury.swin.edu.au/cos10005/s1234567/lab05/surveyhome.html

Note: You can copy the URLs in the table, but remember to replace the unit codes and student id in the above examples with yours to obtain the URLs of your webpages on Mercury.

[IMPORTANT] When the browser authorisation request dialogue pops up, use your SIMS username and password to confirm access.

Step 9: HTML and CSS Validation

To validate the HTML file, use the validator at <http://validator.w3.org>

To validate the CSS file, use the CSS validator at <http://jigsaw.w3.org/css-validator/>